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C Programming Project Presentation

- Project Title: Contact Book(No files version)
- Course: C Programming Lab
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# Introduction

- The Contact Book is a simple C program designed to store and manage personal contact information such as names, phone numbers, and email addresses.
- This project helps in understanding how data can be stored, accessed, and manipulated using arrays and structures in C programming without involving file handling.
- It provides a menu-driven interface that allows users to add, search, delete, and display contacts easily.

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# Objective

- To design a contact management system using C programming.
- To perform operations such as adding, searching, deleting, and displaying contacts.
- To understand the use of arrays and structures for data storage.
- To build a user-friendly menu-driven console application.

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# Problem Statement

- Managing personal or professional contact details manually is time-consuming and prone to error.
- There is a need for a simple computerized contact management system that allows users to quickly store and access contact information
- . This project solves that problem by developing a console-based contact book using basic data structures and C programming concepts.

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# System Requirements

- Hardware requirements:
- processor:Intel i3 or above
- RAM:2GB minimum
- Storage:100 MB free space
- Software requirements :
- Operating System: Windows / Linux
- IDE / Compiler: Code::Blocks, Turbo C, or GCC Compiler

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# System Design

Design Overview:

The system follows a modular design where each function performs a specific task:

- Add Contact
- Display Contacts
- Search Contact
- Delete Contact

Data Structure Used:

A struct is used to represent a contact, and an array stores multiple contacts.

```
struct Contact {
  char name[50];
  char phone[15];
  char email[50];
};
```

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# Module Description

- Add Contact- Allows the user to enter and save contact details.
- Display Contacts- Shows all saved contacts on the screen.
- Search Contact -Searches for a contact by name.
- Delete Contact -Removes a contact from the list by name.
- Exit Ends -the program execution.

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# Algorithm

- Start
- Display menu with options (Add, Display, Search, Delete, Exit)
- Read user choice
- If “Add”, read and store contact details
- If “Display”, show all contacts
- If “Search”, compare entered name with stored names
- If “Delete”, find and remove the contact
- Repeat until user chooses “Exit”
- Stop

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# Implementation

```
#include <stdio.h>
#include <string.h>

struct Contact {
 char name[50];
 char phone[15];
 char email[50];
};

int main() {
struct Contact contacts[100];
int count = 0, i, choice;
char searchName[50];

while (1) {
printf("\n--- Contact Book ---\n");
printf("1. Add Contact\n");
printf("2. Display Contacts\n");
printf("3. Search Contact\n");
printf("4. Delete Contact\n");
printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
getchar(); // to clear input buffer
switch (choice) {
case 1:
printf("Enter name: ");
gets(contacts[count].name);
printf("Enter phone: ");
gets(contacts[count].phone);
printf("Enter email: ");
gets(contacts[count].email);
count++;
printf("Contact added successfully!\n");
break;

case 2:
printf("\n--- Contact List ---\n");
for (i = 0; i < count; i++) {
printf("%d. %s | %s | %s\n", i + 1, contacts[i].name, contacts[i].phone,
contacts[i].email);
}
break;

case 3:
printf("Enter name to search: ");
gets(searchName);
for (i = 0; i < count; i++) {
if (strcmp(searchName, contacts[i].name) == 0) {
printf("Contact Found: %s | %s | %s\n", contacts[i].name, contacts[i].phone,
contacts[i].email);
break;
```

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# Implementation

```
}
}
if (i == count)
printf("Contact not found!\n");
break;

case 4:
printf("Enter name to delete: ");
gets(searchName);
for (i = 0; i < count; i++) {
if (strcmp(searchName, contacts[i].name) == 0) {
for (int j = i; j < count - 1; j++) {
contacts[j] = contacts[j + 1];
}
count--;
printf("Contact deleted successfully!\n");
break;
}
}
if (i == count)
printf("Contact not found!\n");
break;

case 5:
printf("Exiting program...\n");
return 0;

default:
printf("Invalid choice! Try again.\n");
}
}
}
```

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# Output Screens

- SAMPLE OUTPUT

- --- Contact Book ---
- 1. Add Contact
- 2. Display Contacts
- 3. Search Contact
- 4. Delete Contact
- 5. Exit

- Enter your choice: 1
- Enter name: Ramesh
- Enter phone: 9876543210
- Enter email: ramesh@gmail.com
- Contact added successfully!

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# Results & Discussion

- The Contact Book system successfully performs all required operations without using files.
- It uses arrays and structures to manage contacts temporarily during program execution
- The project demonstrates core concepts like arrays, structures, string manipulation, and switch-case control.

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# Conclusion and Future Enhancement

- The Contact Book project helps beginners understand how to handle structured data in C.
- Future enhancements can include:
- Using file handling to permanently save contacts.
- Adding sorting and editing features.
- Creating a graphical user interface (GUI).

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# References / Acknowledgement

- “Let Us C” by Yashavant Kanetkar
- Online C programming documentation (www.programiz.com , www.geeksforgeeks.org )
- Special thanks to our faculty and lab instructor for their guidance and support.

# Thank You

- The Contact Book project is a simple yet powerful example of how structured programming in C can be used to solve real-life problems effectively. It demonstrates how logic, organization, and a clear approach can make data handling smooth even without advanced features like file handling

- Project Motto:
- “Organize. Access. Simplify.”

- This project stands on the principle that even simple tools, when well-designed, can make life easier. A small codebase with clear logic can create meaningful impact in data organization and management..